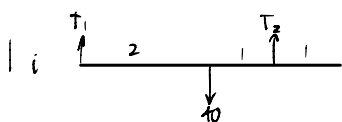


Moments | HW

1/19/26



$$2T_1 = T_2$$

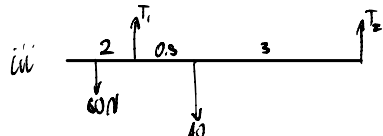
$$T_1 + T_2 = 40$$

$$2(40 - T_2) = T_2$$

$$80 - 2T_2 = T_2$$

$$T_2 = \frac{80}{3}$$

$$\therefore T_1 = \frac{40}{3}$$



$$\frac{1}{2}T_1 - \frac{5}{2}(80) - 3T_2 = 0$$

$$T_1 + T_2 = 120$$

$$T_2 = 120 - T_1$$

$$\therefore \frac{1}{2}T_1 - 200 - 360 + 3T_1 = 0$$

$$\frac{3}{2}T_1 = 560$$

$$T_1 = 160$$

$$\rightarrow T_2 = -40$$

$$2. |P| = 30$$

$$0.6 \times 20 = 30x$$

$$x = \frac{0.6 \times 20}{30}$$

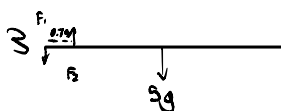
$$= 0.4$$

$$ii. 0.6P + 0.6Q = 30$$

$$30 \times 0.4 = P \times 0.6$$

$$P = 20$$

$$\therefore Q = 10$$

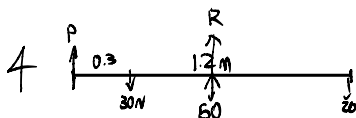


$$F_2 = F_1 + 5g$$

$$0.75F_2 = 2 \times 5g$$

$$F_2 = \frac{2 \times 20}{3} = 13.33 \text{ N}$$

$$F_1 = 81.7 \text{ N}$$



$$0.6P + 0.6 \times 20 = 0.3 \times 30$$

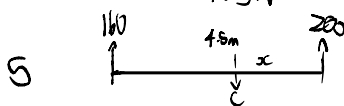
$$P = \frac{0.3 \times 30 - 0.6 \times 20}{0.6}$$

$$= -5 \text{ N}$$

$\therefore 5 \text{ N down}$

$$\therefore R = 50 + 20 + 30 + 5$$

$$= 105 \text{ N}$$



$$C = 360$$

$$200x = 160(4.5 - x)$$

$$360x = 160 \times 4.5$$

$$x = 2 \text{ m}$$

$\therefore$ distance from A is 2.5m

Exam Qs

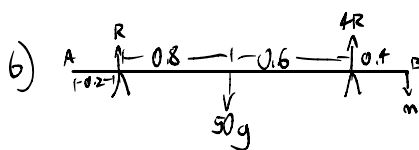
1) a) if magnitude twice as large,

$$0.8C = (1-x)2C$$

$$\frac{0.8}{2} = 1-x$$

$$0.4 = 1-x$$

$$x = 0.6 \text{ m}$$



$$5R = 50g + mg$$

$$0.4 \times 4R + 1.8 \times R = 50g$$

$$R = \frac{50 \times 10}{0.4 \times 4 + 1.8}$$

$$= 144.118$$

$$mg = 5R - 50g$$

$$m = 23.5 \text{ kg}$$